

Python Analytics Api Evidence

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Python Analytics API Evidence

Date: 2026-07-03

Public base URL:

https://academy.example.invalid/api/analytics/v1

Postman collection:

postman/American-Latin-Class-Analytics-API.postman_collection.json

Environment:

postman/American-Latin-Class.postman_environment.json

Objective

This evidence proves the Python API is a real project API, not only an authentication check. The API is implemented with FastAPI and provides academic analytics over the American Latin Class PostgreSQL database.

The API calculates:

- Student attendance risk.
- Student scholarship readiness.
- Branch performance summary.
- Teacher workload and estimated payment.

The same JWT session token created by the Node Auth API is used to protect the analytics endpoints.

Evidence Summary

Evidence	Method	URI	Expected Status	Purpose
Health check	GET	/health	200	Proves the Python FastAPI service is running.
No-token protection	GET	/students/{{student_id}}/attendance-risk	401	Proves protected API routes reject anonymous access.
Attendance risk	GET	/students/{{student_id}}/attendance-risk	200	Calculates attendance percentage and risk level.
Scholarship readiness	GET	/students/{{student_id}}/scholarship-readiness	200	Compares attendance against scholarship rule.
Branch performance	GET	/branches/{{branch_id}}/performance-summary	200	Aggregates branch academic indicators.
Teacher workload	GET	/teachers/{{teacher_id}}/workload-summary	200	Calculates worked hours and estimated pay.
Logout validation	POST + GET	/api/v1/auth/logout, then analytics route	200, then 401	Proves revoked sessions can no longer use the API.

1. Health Check

Postman request:

```
GET {{analytics_base_url}}/health
```

Expected response:

```
{
  "success": true,
  "message": "Analytics API healthy",
  "data": {
    "service": "American Latin Class Analytics API",
    "status": "healthy",
    "authRequired": true
  }
}
```

What this proves:

- The Python API is deployed.
- FastAPI is running behind Nginx.
- Authentication is enabled for protected endpoints.

2. Student Attendance Risk

Postman request:

```
GET {{analytics_base_url}}/students/{{student_id}}/attendance-risk
Authorization: Bearer {{session_token}}
```

Actual verified data:

Field	Value
Student	Camila Rojas REAL-20260624154645
Level	B1
Total records	6
Attended records	6
Absent records	0
Late records	1
Attendance rate	100.0

Field	Value
Risk level	low

Expected response excerpt:

```
{
  "success": true,
  "message": "Student attendance risk",
  "data": {
    "studentName": "Camila Rojas REAL-20260624154645",
    "totalRecords": 6,
    "attendedRecords": 6,
    "attendanceRate": 100.0,
    "riskLevel": "low",
    "recommendation": "Student attendance is healthy."
  }
}
```

How the API works:

```
attendanceRate = attendedRecords / totalRecords * 100
attendanceRate = 6 / 6 * 100 = 100.0
```

Risk rule:

Condition	Risk
No records	no_data
Attendance below 75%	high
Attendance below 90%	medium
Attendance 90% or higher	low

Because the student has 100.0%, the API returns riskLevel: low.

3. Student Scholarship Readiness

Postman request:

```
GET {{analytics_base_url}}/students/{{student_id}}/scholarship-readiness
Authorization: Bearer {{session_token}}
```

Actual verified data:

Field	Value
Attendance rate	100.0
Required attendance rate	90.0
Period months	2
Scholarship eligible	true
Missing percentage points	0

Expected response excerpt:

```
{
  "success": true,
  "message": "Student scholarship readiness",
  "data": {
    "attendanceRate": 100.0,
    "requiredAttendanceRate": 90.0,
    "periodMonths": 2,
    "scholarshipEligible": true,
    "missingPercentagePoints": 0,
    "recommendation": "Student meets the scholarship attendance threshold."
  }
}
```

How the API works:

```
scholarshipEligible = attendanceRate >= requiredAttendanceRate
scholarshipEligible = 100.0 >= 90.0 = true
```

This endpoint proves the API reads the active scholarship rule from the database and applies it to the selected student.

4. Branch Performance Summary

Postman request:

```
GET {{analytics_base_url}}/branches/{{branch_id}}/performance-summary
Authorization: Bearer {{session_token}}
```

Actual verified data:

Field	Value
Branch	Santo Domingo Central REAL-20260624154645

Field	Value
City	Santo Domingo
Students total	6
Students active	6
Teachers total	3
Class groups total	2
Class sessions total	6
Attendance records total	10
Attendance rate	90.0
Pending enrollment requests	5
Performance level	strong

Expected response excerpt:

```
{
  "success": true,
  "message": "Branch performance summary",
  "data": {
    "branchName": "Santo Domingo Central REAL-20260624154645",
    "studentsTotal": 6,
    "teachersTotal": 3,
    "classGroupsTotal": 2,
    "classSessionsTotal": 6,
    "attendanceRecordsTotal": 10,
    "attendanceRate": 90.0,
    "pendingEnrollmentRequests": 5,
    "performanceLevel": "strong"
  }
}
```

How the API works:

- Counts students assigned to the branch.
- Counts active students.
- Counts teachers assigned to the branch.
- Counts class groups and class sessions.
- Aggregates attendance records for branch students.
- Calculates an attendance rate and classifies branch performance.

Performance rule:

Condition	Performance
No attendance records	no_data
Attendance 90% or higher	strong
Attendance 75% to 89.99%	watch
Attendance below 75%	needs_attention

Because this branch has `90.0%`, the API returns `performanceLevel: strong`.

5. Teacher Workload Summary

Postman request:

```
GET {{analytics_base_url}}/teachers/{{teacher_id}}/workload-summary
Authorization: Bearer {{session_token}}
```

Actual verified data:

Field	Value
Teacher	Isabella Torres REAL-20260624154645
Hourly rate	22.5
Check-ins total	1
Completed check-ins	1
Open check-ins	0
Class sessions total	1
Completed hours	3.0
Estimated pay	67.5

Expected response excerpt:

```
{
  "success": true,
  "message": "Teacher workload summary",
  "data": {
    "teacherName": "Isabella Torres REAL-20260624154645",
    "hourlyRate": 22.5,
    "completedHours": 3.0,
    "estimatedPay": 67.5
  }
}
```

```
}  
}
```

How the API works:

```
estimatedPay = completedHours * hourlyRate  
estimatedPay = 3.0 * 22.5 = 67.5
```

This proves the Python API calculates teacher payment projections from database attendance records.

6. Recommended Postman Screenshots

Capture these screens for the final evidence package:

Screenshot	What To Show
1	<code>GET /api/analytics/v1/health</code> with <code>200 OK</code> .
2	<code>GET /students/{{student_id}}/attendance-risk</code> without token returning <code>401</code> .
3	<code>POST /api/v1/auth/login</code> returning <code>200 OK</code> and session token.
4	<code>GET /students/{{student_id}}/attendance-risk</code> with Bearer token returning calculated risk.
5	<code>GET /students/{{student_id}}/scholarship-readiness</code> with scholarship eligibility.
6	<code>GET /branches/{{branch_id}}/performance-summary</code> with branch totals.
7	<code>GET /teachers/{{teacher_id}}/workload-summary</code> with estimated pay.
8	<code>POST /api/v1/auth/logout</code> returning <code>200 OK</code> .
9	Repeat a protected analytics endpoint after logout returning <code>401</code> .

Conclusion

This evidence demonstrates that the Python API:

- Is deployed and reachable through the public HTTPS domain.
- Uses FastAPI as a separate backend API.
- Reads PostgreSQL records from the configured private database; no infrastructure endpoint is published here.
- Performs domain-specific calculations.

- Protects private analytics with the same JWT session system as the Node backend.
- Rejects revoked tokens after logout.